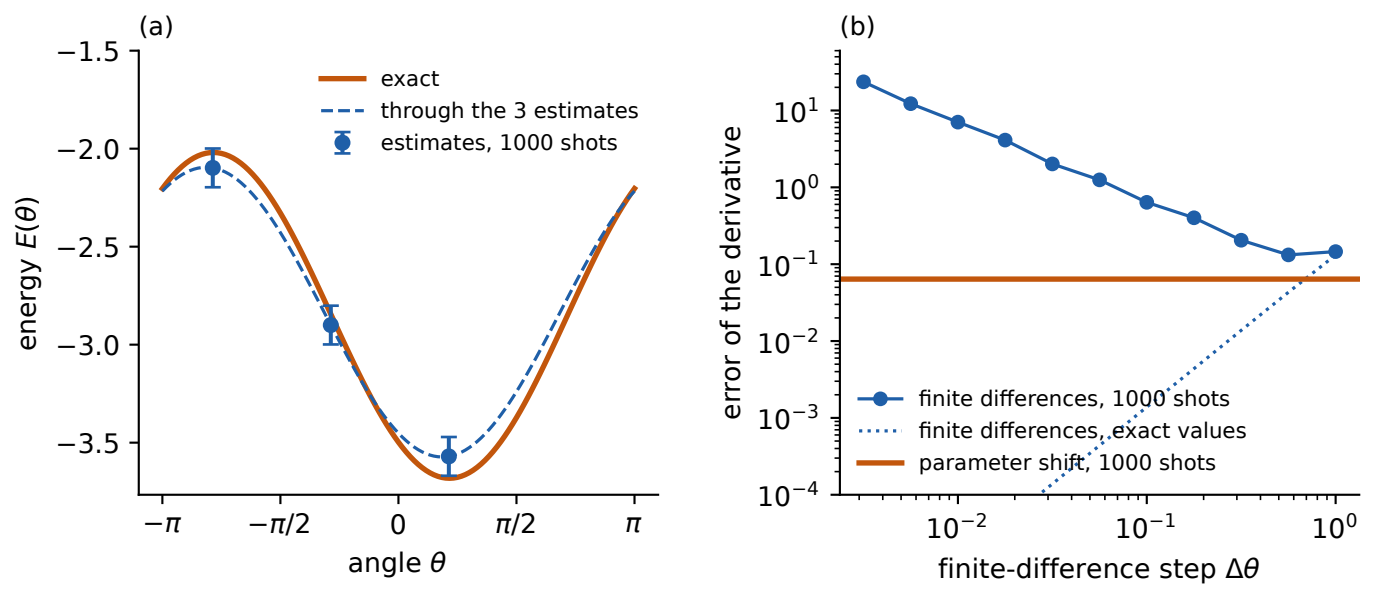
(a)
exact
through the 3 estimates
estimates, 1000 shots
energy $E(\theta)$
angle θ
$-\pi$
$-\pi/2$
0
$\pi/2$
π
−1.5
−2.0
−2.5
−3.0
−3.5
(b)
error of the derivative
finite differences, 1000 shots
finite differences, exact values
parameter shift, 1000 shots
finite-difference step $\Delta\theta$
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{-3}
10^{-4}